Noise Filtering by SVD and SSA

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Outline

Three Main Goals

1. **Estimate the noise level** σ in the data
2. **Estimate the RMSE** (root mean square error) of each SVD mode
3. **Filter noise** by retaining only modes with low RMSE

Data Model

- **Clean data matrix:** $\mathbf{A} \in \mathbb{R}^{T \times D}$
- **Noise matrix:** $\mathbf{N}$ with i.i.d. entries $N_{ij} \sim \mathcal{N}(0, \sigma^2)$
- **Observed data:**
$$\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{N}$$
- **Goal:** Recover $\mathbf{A}$ from $\widetilde{\mathbf{A}}$

Challenges

- True rank r of $\mathbf{A}$ is unknown
- Noise corrupts all singular values and vectors
- Naive truncation risks:
  - Discarding weak but real signal
  - Retaining noise-dominated modes
- **Need:** Data-driven criterion for mode selection

Definition: SVD of Clean Data

Singular Value Decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$

- $\mathbf{U} \in \mathbb{R}^{T \times r}$: left singular vectors
- $\mathbf{S} = \mathrm{diag}(s_1, \ldots, s_r)$: singular values
- $\mathbf{V} \in \mathbb{R}^{D \times r}$: right singular vectors
- $r = \mathrm{rank}(\mathbf{A})$

Mode-k expansion:

$$\mathbf{A} = \sum_{k=1}^{r} s_k \mathbf{u}_k \mathbf{v}_k^T$$

Theorem: Eckart-Young

Best Rank-r Approximation

$$\mathbf{A}_r = \sum_{k=1}^{r} s_k \mathbf{u}_k \mathbf{v}_k^T$$

Error:

$$\|\mathbf{A} - \mathbf{A}_r\|_F^2 = \sum_{k=r+1}^{\mathrm{rank}(\mathbf{A})} s_k^2$$

- $\mathbf{A}_r$ is the best rank-r approximation to $\mathbf{A}$ in Frobenius norm

SVD of Noisy Data

- **Noisy data:** $\widetilde{\mathbf{A}} = \bar{\mathbf{U}}\bar{\mathbf{S}}\bar{\mathbf{V}}^T$
- Due to noise, $\bar{s}_k \neq s_k$
- Noise inflates singular values, especially for higher modes

Low-Rank Reconstruction

- **Reconstruction:**

$$\bar{\mathbf{A}}_r = \sum_{k=1}^{r} \bar{s}_k \bar{\mathbf{u}}_k \bar{\mathbf{v}}_k^T$$

- **Reconstruction loss:**

$$\Delta_r = \|\mathbf{A} - \bar{\mathbf{A}}_r\|_F^2$$

- Δ_r includes both truncation error and noise error

Motivation: Why Use a Hankel Matrix?

- **1D signals** (e.g., time series) are not directly suitable for SVD, which operates on matrices.
- **Hankel embedding** transforms a 1D signal into a structured 2D matrix, preserving temporal relationships.
- Enables:
  - Extraction of trends, oscillations, and periodicities
  - Effective separation of signal and noise
  - Application of SVD/PCA techniques
- Central to **Singular Spectrum Analysis (SSA)** and Hankel-SVD methods.

Hankel Matrix Construction

- Given a 1D signal $s = (s_1, s_2, \ldots, s_N)$ of length N
- Choose a window length L $(2 < L \leq N/2)$
- Number of columns: $K = N - L + 1$
- **Hankel matrix:**

$$H_{i,j} = s_{i+j-1}, \quad 1 \leq i \leq L, \quad 1 \leq j \leq K$$

- All elements along each anti-diagonal are equal

Example:

$$\mathbf{H} = \begin{bmatrix} s_1 & s_2 & \cdots & s_K \\ s_2 & s_3 & \cdots & s_{K+1} \\ \vdots & \vdots & \ddots & \vdots \\ s_L & s_{L+1} & \cdots & s_N \end{bmatrix}$$

SVD of the Hankel Matrix

- Apply SVD to the Hankel matrix:

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

  - $\mathbf{U}$: left singular vectors (temporal patterns)
  - $\mathbf{\Sigma}$: singular values (mode strengths)
  - $\mathbf{V}$: right singular vectors (lagged patterns)
- **Truncated SVD:** Keep only the first r dominant modes to suppress noise
- **Low-rank approximation:**

$$\mathbf{H}_r = \sum_{k=1}^{r} \sigma_k \mathbf{u}_k \mathbf{v}_k^T$$

Signal Reconstruction: Diagonal Averaging

- After SVD and truncation, reconstruct the denoised signal by **diagonal averaging** (Hankelization)
- For each anti-diagonal $d = i + j - 1$, average all $H_{i,j}$ with the same d
- The reconstructed signal $\hat{s}_n$ is:

$$\hat{s}_n = \frac{1}{\#\{(i,j) : i + j - 1 = n\}} \sum_{i+j-1=n} H_{i,j}$$

  for $n = 1, 2, \ldots, N$
- This step ensures the output is a 1D time series of the original length

Choosing the Window Length L

- L is a critical parameter:
  - Too small: poor separation of signal and noise
  - Too large: insufficient columns, increased error
- **Guidelines:**
  - $2 < L \leq N/2$
  - Common choices: $L \approx N/4$ or $L \approx (N+1)/2$
- Optimal L enhances the ability to distinguish signal components from noise

Connection to Singular Spectrum Analysis (SSA)

- **SSA** is a nonparametric method for time series analysis based on:
  1. Hankel embedding of the 1D signal
  2. SVD of the Hankel matrix
  3. Grouping and truncation of SVD modes
  4. Reconstruction via diagonal averaging
- Applications:
  - Denoising
  - Trend and periodicity extraction
  - Forecasting and missing data imputation
- SSA and Hankel-SVD are widely used in signal processing, economics, and biomedical engineering

Singular Spectrum Analysis (SSA)

Singular Spectrum Analysis (SSA) is a non-parametric technique for time-resolved signals analysis and decomposition. It is used to decompose a time series into several interpretable components: a trend, oscillatory components (periodic or quasi-periodic), and noise. Unlike classical methods like ARIMA, SSA does not require a prior model for the data-generating process, making it highly effective for analyzing complex, non-stationary signals. It consists of the following steps:

Step 1: Embedding

- Given a time series $\{x_t\}_{t=1}^{N}$, construct a trajectory matrix $\mathbf{X}$ using a sliding window of length L:

$$\mathbf{X} = \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ x_2 & x_3 & \cdots & x_{K+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_L & x_{L+1} & \cdots & x_N \end{bmatrix},$$

  where $K = N - L + 1$.
- $\mathbf{X} \in \mathbb{R}^{L \times K}$ is a Hankel matrix, meaning all elements along its anti-diagonals are equal.

Step 2: Singular Value Decomposition (SVD)

- Perform the Singular Value Decomposition (SVD) of the trajectory matrix $\mathbf{X}$:

$$\mathbf{X} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top,$$

  where:
  - σ_i are the singular values, ordered such that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$,
  - $\mathbf{u}_i$ are the left singular vectors (principal components),
  - $\mathbf{v}_i$ are the right singular vectors.
- The rank r of $\mathbf{X}$ is determined by the number of non-zero singular values.

Step 3: Grouping

- Group the singular values and corresponding singular vectors into subsets to form groups of components:

$$\mathbf{X} = \mathbf{X}_1 + \mathbf{X}_2 + \cdots + \mathbf{X}_m,$$

  where $\mathbf{X}_i = \sum_{j \in I_i} \sigma_j \mathbf{u}_j \mathbf{v}_j^\top$ and I_i is the index set for group i.
- This step allows separation of the signal into meaningful components, such as trends, oscillatory patterns, and noise.

Step 4: Diagonal Averaging (Hankelization)

- Reconstruct the time series from each grouped matrix $\mathbf{X}_i$ by averaging along the anti-diagonals:

$$\hat{x}_t = \frac{1}{\text{number of elements in diagonal}} \sum_{\text{diagonal elements}} \mathbf{X}_i.$$

- The final reconstructed time series is obtained by summing the contributions from all groups:

$$\hat{x}_t = \hat{x}_t^{(1)} + \hat{x}_t^{(2)} + \cdots + \hat{x}_t^{(m)}.$$

Key Features of SSA

- Non-parametric: No assumptions about the underlying model of the time series.
- Decomposition: Separates the time series into trend, oscillatory components, and noise.
- Applications: Used in signal processing, climate studies, economics, and more.
- Comparison with SVD Filtering:
  - Both techniques use SVD, but SSA explicitly reconstructs the time series using diagonal averaging.
  - SSA is tailored for time series analysis, while SVD filtering is more general.

Hankel Matrix Approach

- **Transforms** 1D time series into a 2D matrix for SVD
- **Enables** separation of signal and noise via low-rank approximation
- **Reconstruction** by diagonal averaging yields a denoised signal
- **Central** to SSA and related time series analysis methods

Summary & Conclusions

- Principled SVD-based and SSA-based noise filtering for experimental data
- Explicit estimation of noise variance from data
- RMSE-based criterion for mode selection
- Analytic estimate of reconstruction accuracy
- Outperforms empirical and fixed-threshold approaches

Does it make sense?

Questions & Discussion